

Assignment: Cardiology Coronary Artery Image Processing

The system should be able to accurately process **cardiology X-ray/cine images of the heart and coronary arteries** to improve the visibility of contrast-filled blood vessels. The system must suppress unwanted anatomical background structures such as **ribs, spine, lungs, and other background noise** while preserving the coronary artery structures.

The application should support **16-bit grayscale cardiology images** and handle variations in image quality, contrast, noise, motion, and anatomical structures. It must apply appropriate image preprocessing and background-subtraction techniques to enhance the visibility of coronary arteries without losing important vascular information.

A key requirement is that the image-processing pipeline must operate with **very low latency**, with the **maximum processing time limited to 36 milliseconds per frame**. This latency requirement is **mandatory** and should be demonstrated and measured during the project evaluation.

The processed output should clearly display the coronary arteries with the unwanted background structures sufficiently suppressed, allowing users to visually compare the **original/raw image and processed image**.

Deliverables

Assignment must include:

- A **fully functional codebase** for cardiology X-ray/cine image processing.
- **Dataset(s)** containing suitable cardiology/coronary artery images or legally usable/synthetic datasets for development and testing.
- **Trained model file(s)**, if a deep-learning-based approach is used.
- Image-processing pipeline for:
 - **Background noise suppression**
 - **Rib suppression/removal**
 - **Spine suppression/removal**
 - **Lung/background structure suppression**
 - **Coronary artery enhancement**
 - **Contrast and image-quality enhancement**
- Support for **16-bit grayscale images**.

- The system should preserve important coronary artery information while suppressing unwanted anatomical structures.
- The system should provide a comparison of:
 - **Original/Raw Image**
 - **Processed Image**
 - **Enhanced Coronary Artery Image**
- **Processing latency must be ≤ 36 milliseconds per frame.**
- The project must include **latency measurement/benchmarking** demonstrating that the processing requirement is achieved.
- Complete **project documentation** explaining the implemented approach, algorithms/models, preprocessing, processing pipeline, and performance.
- A README.md file with:
 - Setup instructions
 - Installation instructions
 - Usage guidelines
 - Dataset information
 - Model information
 - Processing pipeline explanation
 - Performance/latency results
- A requirements.txt file listing all dependencies.

The projects should be **deployed on GitHub** for demonstration purposes (recommended).

Performance Requirement

The system must be optimized for **real-time cardiology image processing**.

Maximum permitted processing latency: 36 ms per frame

The latency should be measured from the input frame received by the processing pipeline until the final processed frame is produced.

The evaluation should report:

- Average processing latency

- Minimum processing latency
- Maximum processing latency
- Frames Per Second (FPS)
- Hardware configuration used for testing

The 36 ms latency requirement is compulsory.

Expected Output

The final system should demonstrate that unwanted anatomical structures such as **ribs, spine, lungs, and background noise are suppressed**, while the **coronary arteries remain clearly visible and enhanced**.

The solution should prioritize:

Accuracy + Coronary Vessel Visibility + Background Suppression + Real-Time Performance

Deadline: 9th September 2026 till 9:00 AM

All the Best!